

Responsibility of Software Development in Autonomous Vehicle Accidents

An Engineering Ethics Perspective

Introduction

- Autonomous vehicles are transforming transportation.
- Software is central to decision-making and safety.
- Ethical dilemma: Who is responsible when software fails?

The Ethical Challenge

- Responsibility involves developers, engineers, managers, policymakers.
- Failure sources: Bugs, sensor misinterpretation, incomplete testing.
- Ethical accountability is complex in AV systems.

Engineering Ethics Overview

- Commitment to safety, honesty, public welfare.
- IEEE, NSPE, ACM ethical codes guide decisions.
- Critical for software in safety-critical systems.

Case Study Examples

- Uber AV incident (2018): Software failed to identify pedestrian.
- Tesla Autopilot accidents: Incomplete autonomy and over-reliance.

Key Ethical Principles

- Public safety and welfare.
- Accountability and transparency.
- Informed consent and user education.
- Honesty in data reporting.
- Moral autonomy of engineers.

Types of Responsibilities

- Technical: Code correctness, testing.
- Moral: Anticipate harms.
- Legal: Liability for defects.
- Social: Trust and communication.

Responsibility Distribution

- Developers: Software integrity.
- Designers: Safe human-machine interaction.
- Managers: Resource allocation for safety.
- Corporations: Ethical policy enforcement.
- Governments: Regulations and standards.

Moral Crumple Zone Problem

- Engineers blamed for failures in complex systems.
- Ethical dilemma: Is it fair to blame individuals?

Ethical Software Lifecycle

- Ethical design: Safety over cost.
- Diverse scenario testing.
- Continuous monitoring post-deployment.
- Transparent user communication.

Risk Management in AV Software

- Implement fail-safes and redundancies.
- Ethical algorithms for crash dilemmas.
- Mitigate bias in AI models.

Legal and Regulatory Framework

- Current gaps in legislation.
- Need for transparent audit trails.
- Industry-wide safety standards.
- Clear liability guidelines.

Professional Codes and Guidance

- IEEE/ACM Code: Public interest, disclose risks.
- NSPE Code: Protect public safety and welfare.

Towards Ethical AI in AVs

- Adopt Explainable AI (XAI).
- Promote interdisciplinary collaboration.
- Develop ethical AI governance.

Recommendations

- Include ethics in software engineering education.
- Conduct ethical audits for AV software.
- Foster partnerships for safety regulation.

Conclusion

- Ethical engineering is essential for AVs.
- Balance innovation with public responsibility.
- Continuous ethical reflection builds trust.

Q&A

- Open discussion: Liability, ethical coding, future challenges.